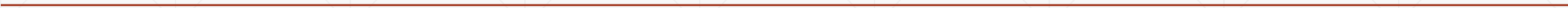


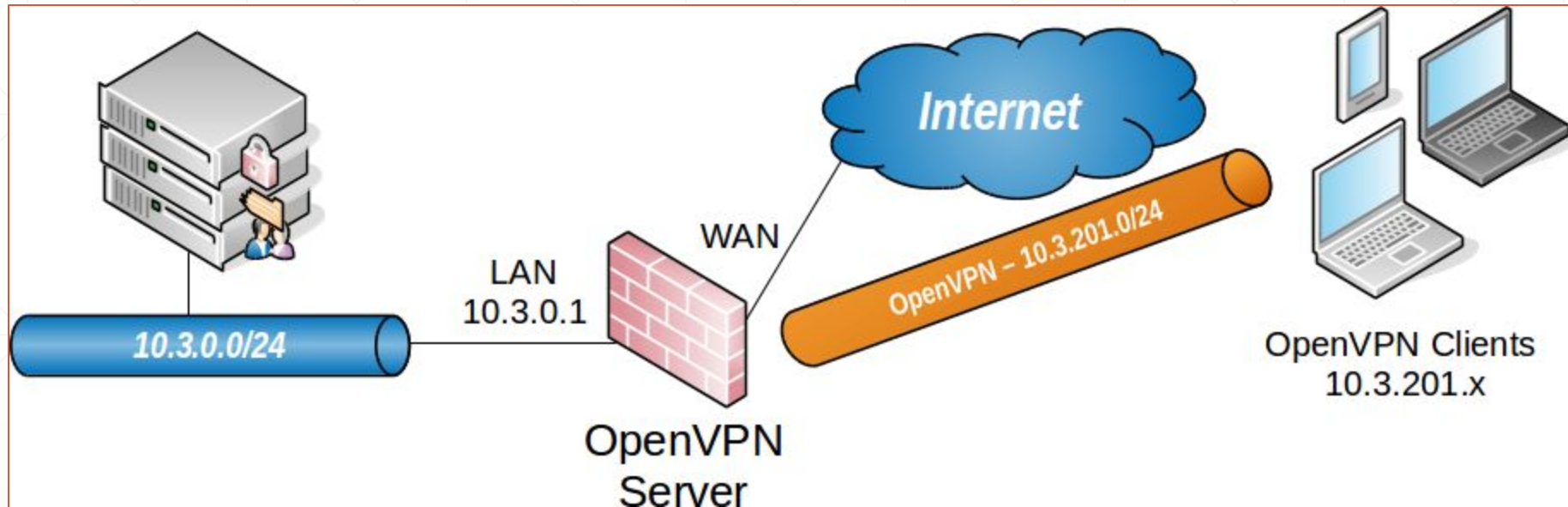
PHẦN 6

KHAI BÁO REMOTE ACCESS VPN VỚI OPENVPN



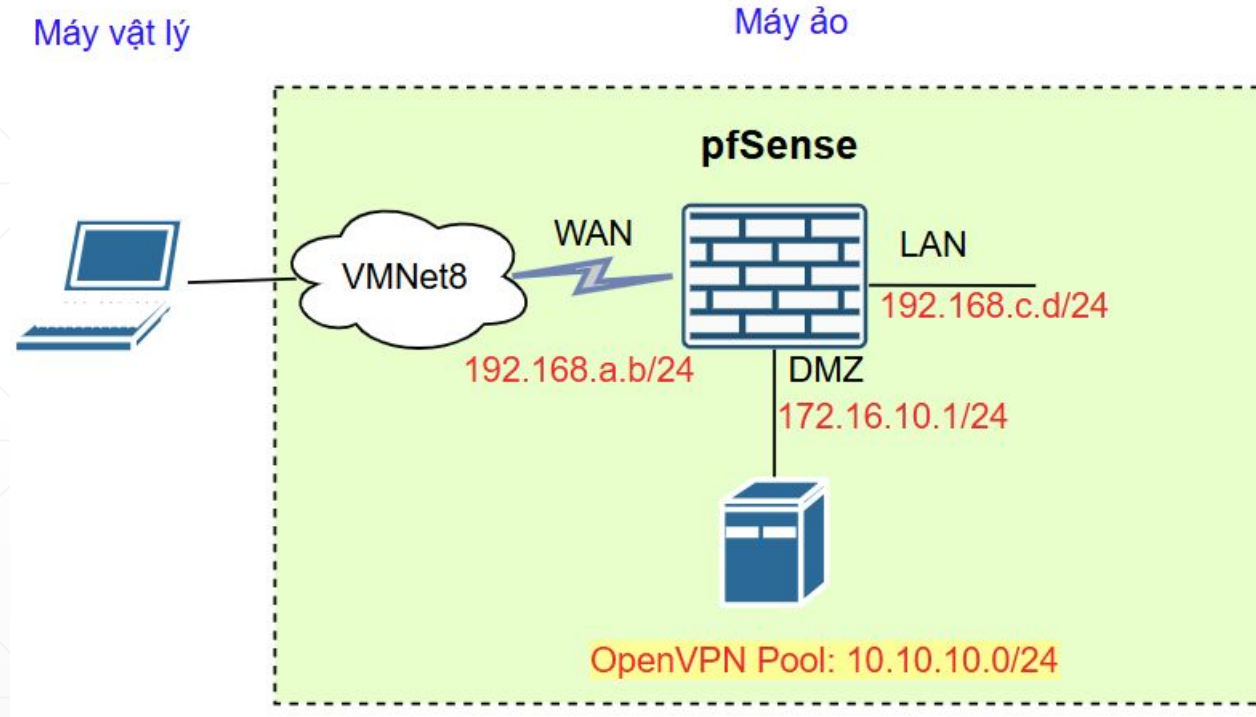
Tổng quan về OpenVPN

- OpenVPN là một giao thức VPN mã nguồn mở chạy độc lập trên nền Linux.
- OpenVPN được tích hợp sẵn trong pfSense mà không cần phải cài đặt thêm và được sử dụng cho Remote VPN (Client to Site).
- OpenVPN Client có thể được cài trên Windows, MAC, Linux, Android, IOS.



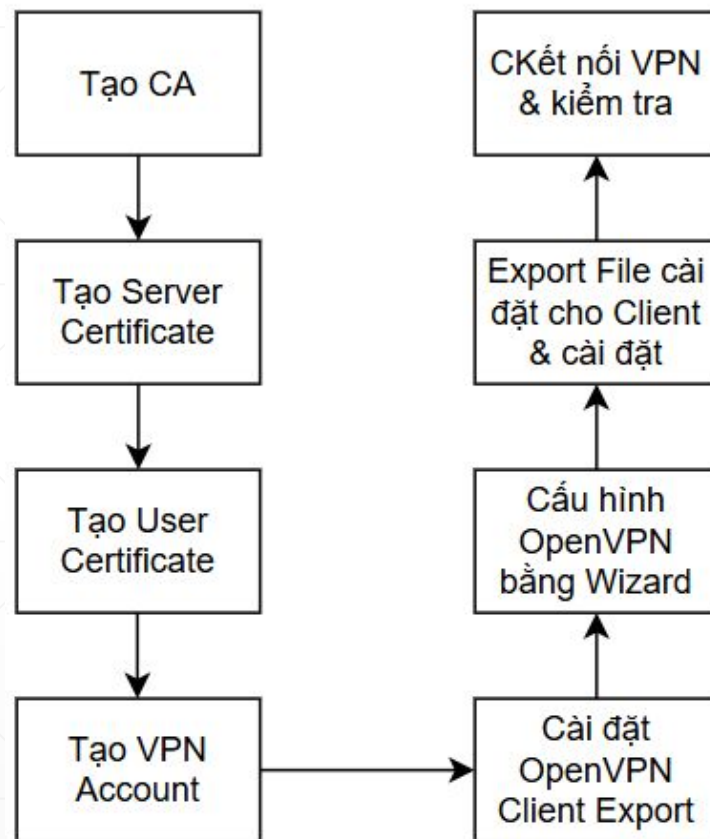
Sơ đồ LAB

- pfSense được cài trên máy ảo, có thể thêm một máy ảo làm Web Server để kiểm tra
- OpenVPN Client có thể được cài trên máy vật lý để kết nối VPN
- Địa chỉ LAN/WAN theo thực tế của VMWare
- IP Pool cấp phát cho VPN Client là 10.10.10.0/24



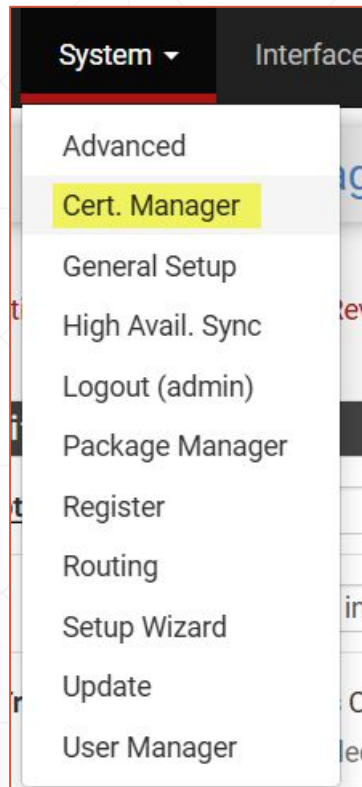
Trình tự thực hiện

- Các bước thực hiện theo trình tự bên dưới.



Tạo CA

- pfSense cho phép tạo CA dùng nội bộ (Self-signed)



System / Certificate / Authorities / Edit

Authorities Certificates Revocation

Create / Edit CA

<u>Descriptive name</u>	VKU_CA
	The name of this entry as displayed in the web interface. This name can contain spaces but it cannot contain special characters.
<u>Method</u>	Create an internal Certificate Authority
<u>Trust Store</u>	☐ Add this Certificate Authority to the system trust store. When enabled, the contents of the CA will be added to the system trust store.
<u>Randomize Serial</u>	☐ Use random serial numbers when signing. When enabled, if this CA is capable of signing, the serial numbers will be checked for uniqueness instead of using the current serial number.

Internal Certificate Authority

<u>Key type</u>	RSA
	2048
	The length to use when generating a new key. The Key Length should not be lower than 1024 bits.

<u>Lifetime (days)</u>	3650
<u>Common Name</u>	internal-ca
	The following certificate authority subject will be used to generate the certificate.
<u>Country Code</u>	VN
<u>State or Province</u>	DaNang
<u>City</u>	DaNang
<u>Organization</u>	VKU
<u>Organizational Unit</u>	Computer Engineering & Electronics

Tạo Server Certificate

- Tạo Server Certificate để xác thực với Client

Authorities **Certificates** Certificate Revocation

Add/Sign a New Certificate

Method

Create an internal Certificate

Descriptive name

VKU VPN Server Certificate

The name of this entry as displayed in the console.
This name can contain spaces but it cannot contain special characters.

Internal Certificate

Certificate authority

VKU_CA

Key type

RSA

2048

The length to use when generating a new key.
The Key Length should not be lower than 1024 bits.

Digest Algorithm

sha256

The digest method used when the certificate is signed.
The best practice is to use SHA256 or higher.
Older algorithms are considered invalid.

Lifetime (days)

3650

Common Name

VKU_CERT

The following certificate subject components are displayed:

Country Code

VN

State or Province

DaNang

City

DaNang

Organization

VKU

Organizational Unit

Computer Engineering & Electronics

Certificate Attributes

Attribute Notes

The following attributes are added to certificate in selected mode.

For Internal Certificates, these attributes are added.

Certificate Type

Server Certificate

Add type-specific usage attributes to the signed certificate.

Alternative Names

FQDN or Hostname

Tạo User Certificate

- Tạo User Certificate để xác thực với Server

User Certificates

Name	CA

+ Add

>> Move to "Member of" list

Hold down CTRL (PC)/COMMAND (Mac) key to select multiple items

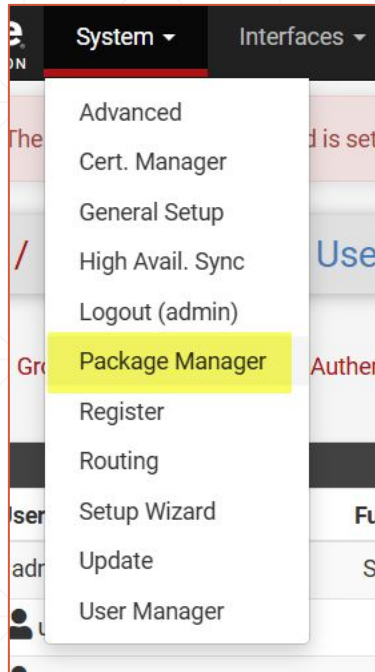
Certificate	☒ Click to create a user certificate
-------------	--

Create Certificate for User

Descriptive name	vpn01
Certificate authority	VKA_CA
<u>Key type</u>	RSA
	2048
	The length to use when generating a new RSA key, in bits. The Key Length should not be lower than 2048 or some
<u>Digest Algorithm</u>	sha256
	The digest method used when the certificate is signed. The best practice is to use an algorithm stronger than

Cài đặt OpenVPN Client Export

- Cài đặt Package



System / Package Manager / Available Packages

Installed Packages Available Packages

Search

Search term Both

Enter a search string or *nix regexp

Packages

Name	Version	Description
openvpn-client-export	1.9.2	Exports pre-configured OpenVPN Client Export

Package Dependencies:
[openvpn-client-export-2.6.7](#)

Please wait while the installation of **pfSense-pkg-openvpn-client-export** completes.
This may take several minutes. Do not leave or refresh the page!

Installed Packages Available Packages **Package Installer**

Package Installation

All repositories are up to date.
The following 5 package(s) will be affected (of 0 checked):

New packages to be INSTALLED:

- 7-zip: 23.01 [pfSense]
- libsysinfo: 0.0.3_2 [pfSense]
- openvpn-client-export: 2.6.7 [pfSense]
- pfSense-pkg-openvpn-client-export: 1.9.2 [pfSense]
- zip: 3.0_1 [pfSense]

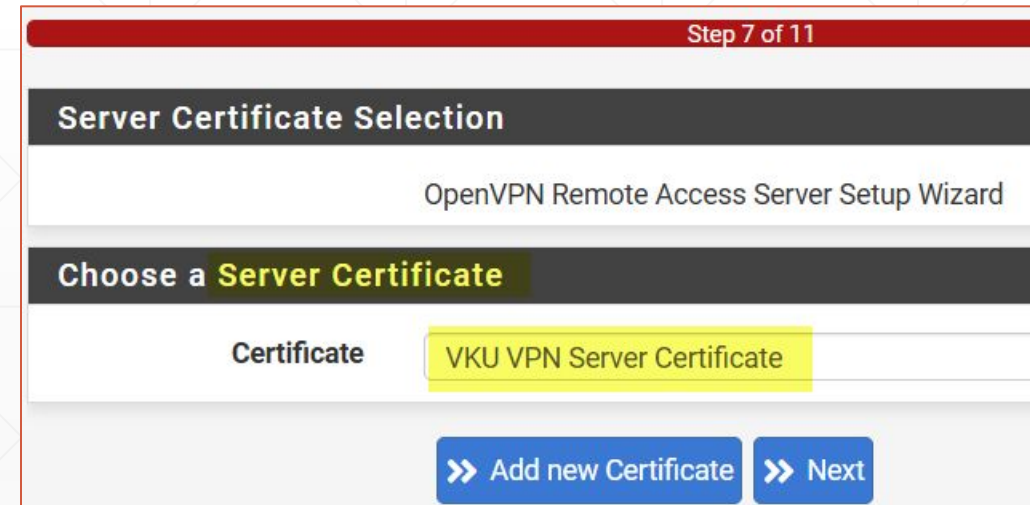
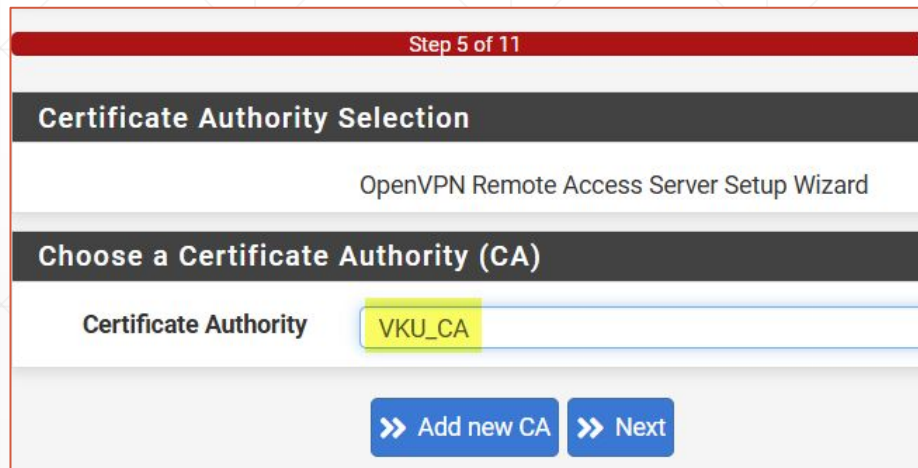
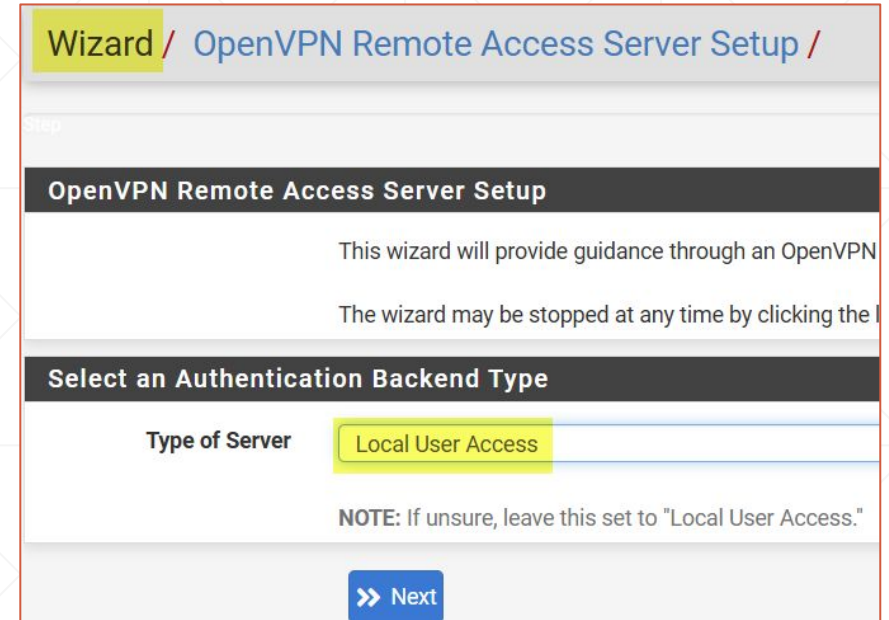
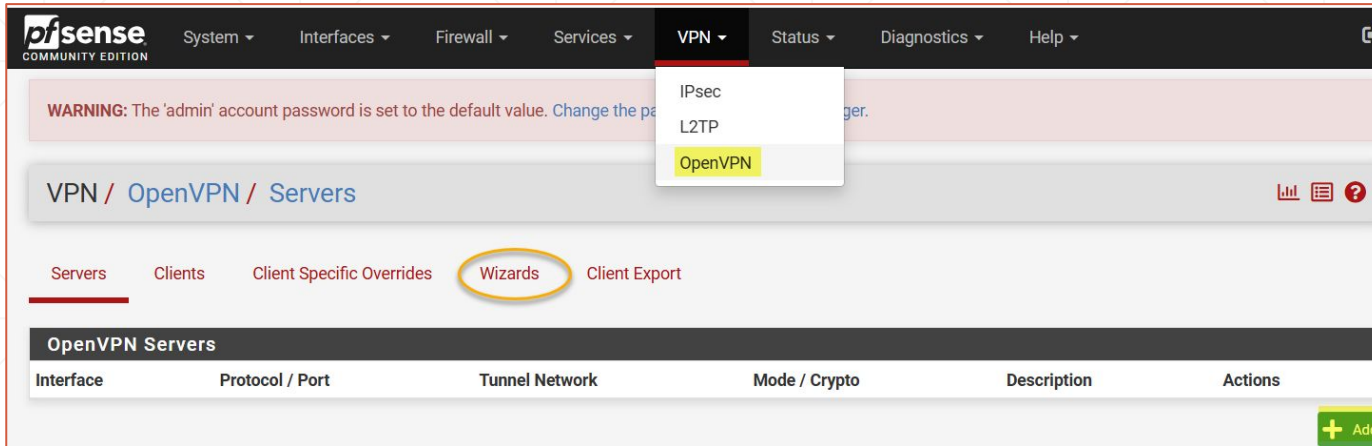
Number of packages to be installed: 5

The process will require 31 MiB more space.
23 MiB to be downloaded.

[1/5] Fetching openvpn-client-export-2.6.7.pkg:

Cấu hình OpenVPN

- Chúng ta sử dụng Wizard



Cấu hình OpenVPN (tt)

Server Setup

OpenVPN Remote Access Server Setup Wizard

General OpenVPN Server Information

Description VKU LAB_Remote VPN
A name for this OpenVPN instance, for administrative service (e.g. "Remote Technical Staff"). It is also

Endpoint Configuration

Protocol UDP on IPv4 only
Protocol to use for OpenVPN connections. If un

Interface WAN
The interface where OpenVPN will listen for inco

Local Port 1194
Local port upon which OpenVPN will listen for co

Tunnel Settings

IPv4 Tunnel Network 10.10.10.0/24
This is the virtual network used for private commu
first network address will be assigned to the serve

Redirect IPv4 Gateway ☐ Force all client generated traffic through the tun

IPv4 Local Network 192.168.74.0/24
This is the network that will be accessible from the
local network through this tunnel on the remote ma

Concurrent Connections
Specify the maximum number of clients allowed to

Allow Compression Refuse any non-stub compression (Most secure)
Allow compression to be used with this VPN instar

Compression Disable Compression [Omit Preference]

Compress tunnel packets using the chosen option
compression is not allowed. Adaptive compression
packets is not being compressed efficiently.

Type-of-Service ☐ Set the TOS IP header value of tunnel packets t

Inter-Client Communication ☐ Allow communication between clients connect

Duplicate Connections ☐ Allow multiple concurrent connections from cli
NOTE: This is not generally recommended, but ma

Firewall Rule Configuration

OpenVPN Remote Access Serv

Rules control passing or blocki

Rules must be added which allo
OpenVPN tunnel.

The options on this step can ad

Traffic from clients to server

Firewall Rule ☒ Add a rule to permit connec

Traffic from clients through VPN

OpenVPN rule ☒ Add a rule to allow all traffic

Finished!

OpenVPN Remote Access Server Setup Wizard

Configuration Complete!

The configuration is now complete.

Adding users for the VPN depends on the chose
For remote authentication servers, add certifica

To easily export client configurations, browse to

>> Finish

Kết quả sau cấu hình OpenVPN

Servers Clients Client Specific Overrides Wizards Client Export

OpenVPN Servers

Interface	Protocol / Port	Tunnel Network	Mode / Crypto	Description	Actions
WAN	UDP4 / 1194 (TUN)	10.10.10.0/24	Mode: Remote Access (SSL/TLS + User Auth) Data Ciphers: AES-256-GCM, AES-128-GCM, CHACHA20-POLY1305, AES-256-CBC Digest: SHA256 D-H Params: 2048 bits	VKU LAB_Remote VPN	  

Floating **WAN** LAN OpenVPN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	
☐	✓	0/0 B	IPv4	UDP	*	*	WAN address	1194 (OpenVPN)	*	none	OpenVPN VKU LAB_Remote VPN wizard

Floating WAN LAN **OpenVPN**

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description
☐	✓	0/0 B	IPv4	*	*	*	*	*	none	OpenVPN VKU LAB_Remote VPN wizard

Tạo Account cho VPN User

System / **User Manager / Users / Edit**

Users Groups Settings Authentication Servers

User Properties

Defined by USER

Disabled ☐ This user cannot login

Username vpn01

Password

Full name OpenVPN user
User's full name, for administrative information only

Expiration date
Leave blank if the account shouldn't expire, otherwise enter the expiration date as MM/DD/YYYY

Custom Settings ☐ Use individual customized GUI options and dashboard layout for this user.

Group membership admins

Certificate ☒ Click to create a user certificate

Create Certificate for User

Descriptive name vnp01 user

Certificate authority VKU_CA

Key type RSA

2048
The length to use when generating a key. The Key Length should not be lower than 1024 bits.

Digest Algorithm sha256
The digest method used when the certificate is generated. The best practice is to use an algorithm that is at least 128 bits long.

Lifetime 3650

Xuất bộ cài đặt cho VPN User

pfSense COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ Services ▾ **VPN ▾** Status ▾

WARNING: The 'admin' account password is set to the default value. [Change the password.](#)

OpenVPN / Client Export Utility

Server Client Client Specific Overrides Wizards Client Export

OpenVPN Server

Remote Access Server VKU LAB_Remote VPN UDP4:1194 ▾

Client Connection Behavior

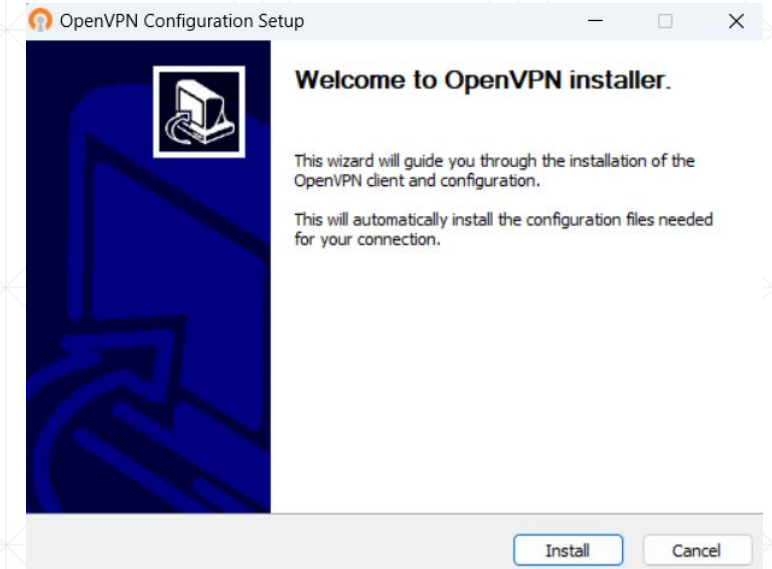
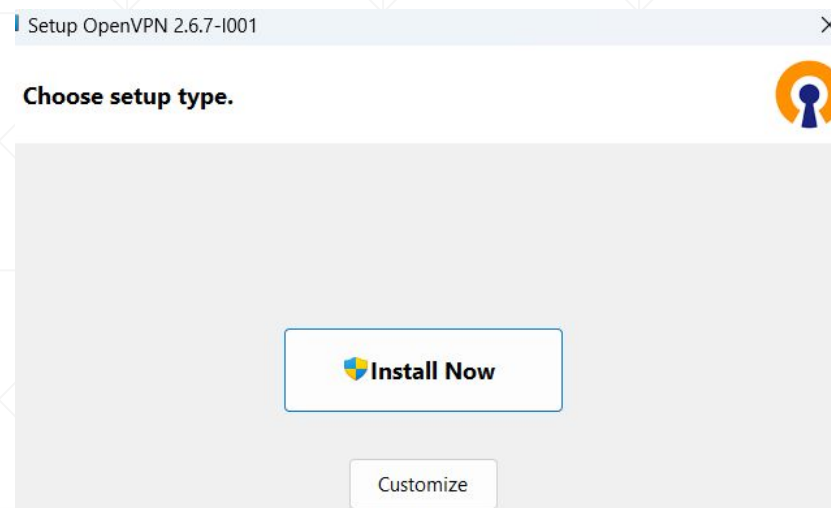
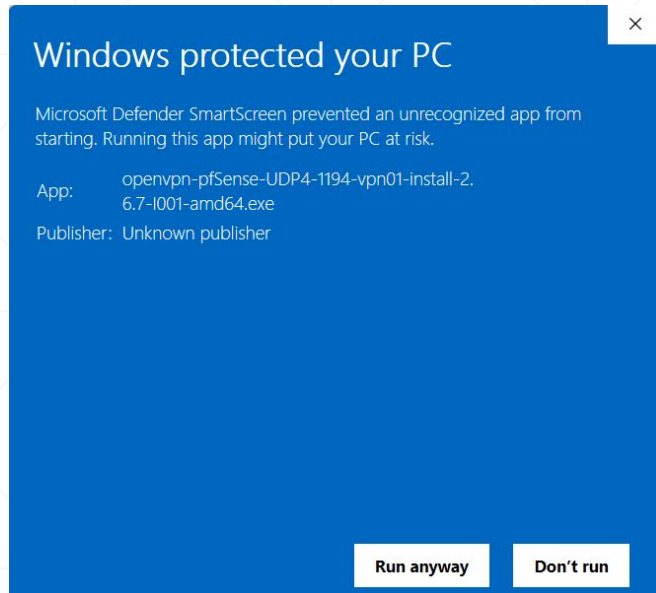
Host Name Resolution Interface IP Address ▾

Verify Server CN Automatic - Use verify-x509-name where possible ▾
Optionally verify the server certificate Common Name (CN) when the client connects

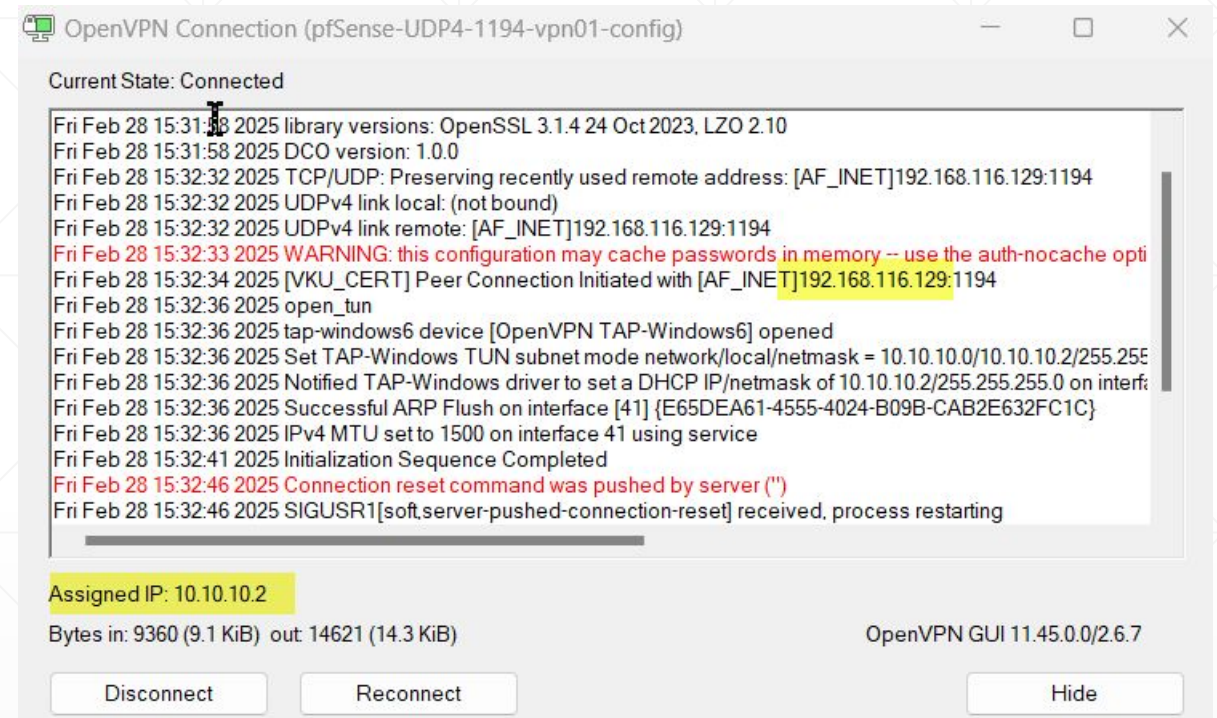
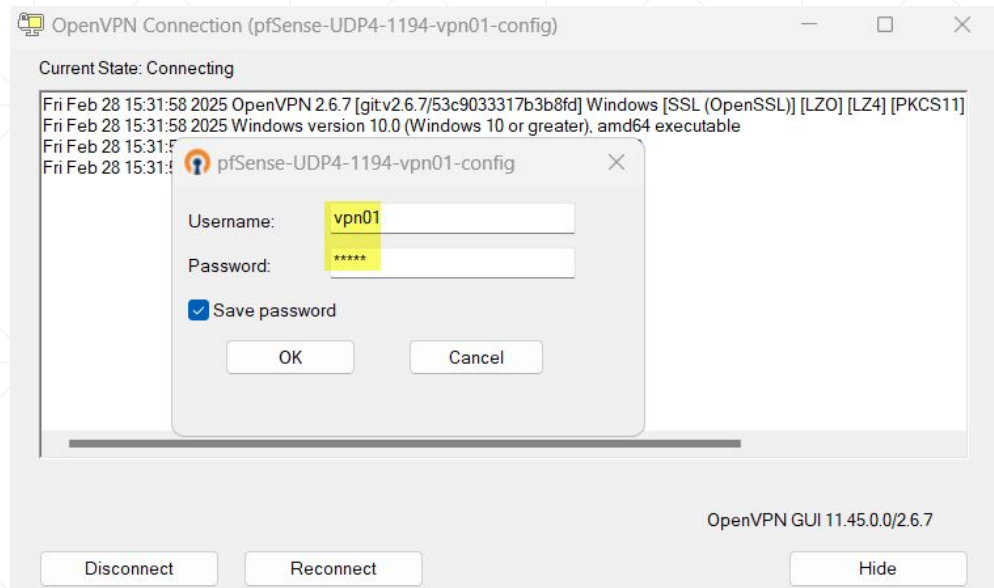
OpenVPN Clients

User	Certificate Name	Export
vpn01	vnp01 user	- Inline Configurations: Most Clients Android OpenVPN Connect (iOS/Android) - Bundled Configurations: Archive Config File Only - Current Windows Installers (2.6.7-lx001): 64-bit 32-bit - Previous Windows Installers (2.5.9-lx601): 64-bit 32-bit - Legacy Windows Installers (2.4.12-lx601): 10/2016/2019 7/8/8.1/2012r2 - Viscosity (Mac OS X and Windows): Viscosity Bundle Viscosity Inline Config

Cài đặt OpenVPN cho Client



Kết nối Remote VPN



Kiểm tra thông tin VPN Client

PN ▾ Status ▾ Diagnosti

ord in th

- Captive Portal
- CARP (failover)
- Dashboard
- DHCP Leases
- DHCPv6 Leases
- DNS Resolver
- Filter Reload
- Gateways
- Interfaces
- IPsec
- Monitoring
- NTP
- OpenVPN**
- Queues
- Services
- System Logs
- Traffic Graph
- UPnP & NAT-PMP

Status / OpenVPN

ovpns1: VKU LAB_Remote VPN UDP4:1194 / Client Connections: 1

Common Name	Real Address	Virtual Address	Last Change	Bytes Sent	Bytes Received	Cipher	Actions
vpn01	192.168.116.1:53766	10.10.10.2	2025-02-28 15:32:53	41 KiB	12 KiB	AES-256-GCM	✕ ✕

Status / Dashboard

Available Widgets

System Information

Name	pfSense.vku.lab
User	admin@10.10.10.2 (Local Database)
System	VMware Virtual Machine Netgate Device ID: bed0a635877f5627197c
BIOS	Vendor: Phoenix Technologies LTD Version: 6.00 Release Date: Thu Nov 12 2020
Version	2.7.2-RELEASE (amd64) built on Thu Dec 7 3:10:00 +07 2023

Interfaces

WAN	↑	1000baseT <full-duplex>	192.168.116.129
LAN	↑	1000baseT <full-duplex>	192.168.74.10

OpenVPN

VKU LAB_Remote VPN UDP4:1194 (1)

Name/Time	Real/Virtual IP	
vpn01	192.168.116.1:53766	✕ ✕
2025-02-28 15:32:53	10.10.10.2	

Kiểm tra kết nối từ VPN Client

```
C:\>route print
```

Interface List

```
34.....Wintun Userspace Tunnel
41...00 ff e6 5d ea 61 .....TAP-Windows Adapter V9
45.....OpenVPN Data Channel Offload
15...f8 9e 94 e8 00 04 .....Microsoft Wi-Fi Direct Virtual Adapter
16...fa 9e 94 e8 00 03 .....Microsoft Wi-Fi Direct Virtual Adapter #2
10...f8 9e 94 e8 00 03 .....Intel(R) Wi-Fi 6E AX211 160MHz
 6...00 50 56 c0 00 01 .....VMware Virtual Ethernet Adapter for VMnet1
18...00 50 56 c0 00 08 .....VMware Virtual Ethernet Adapter for VMnet8
12...f8 9e 94 e8 00 07 .....Bluetooth Device (Personal Area Network)
 1.....Software Loopback Interface 1
```

IPv4 Route Table

Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
10.10.10.0	255.255.255.0	On-link	10.10.10.2	281
10.10.10.2	255.255.255.255	On-link	10.10.10.2	281
10.10.10.255	255.255.255.255	On-link	10.10.10.2	281
127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
127.0.0.1	255.255.255.255	On-link	127.0.0.1	331
127.255.255.255	255.255.255.255	On-link	127.0.0.1	331
169.254.0.0	255.255.0.0	On-link	169.254.33.146	286
169.254.33.146	255.255.255.255	On-link	169.254.33.146	286
169.254.255.255	255.255.255.255	On-link	169.254.33.146	286
172.16.10.0	255.255.255.0	10.10.10.1	10.10.10.2	281
192.168.74.0	255.255.255.0	On-link	192.168.74.1	291
192.168.74.1	255.255.255.255	On-link	192.168.74.1	291

```
C:\>ping 172.16.10.1
```

Pinging 172.16.10.1 with 32 bytes of data:

Reply from 172.16.10.1: bytes=32 time=13ms TTL=64

Reply from 172.16.10.1: bytes=32 time=33ms TTL=64

Reply from 172.16.10.1: bytes=32 time=2ms TTL=64

Reply from 172.16.10.1: bytes=32 time=2ms TTL=64

Ping statistics for 172.16.10.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 33ms, Average = 12ms



Thảo luận
